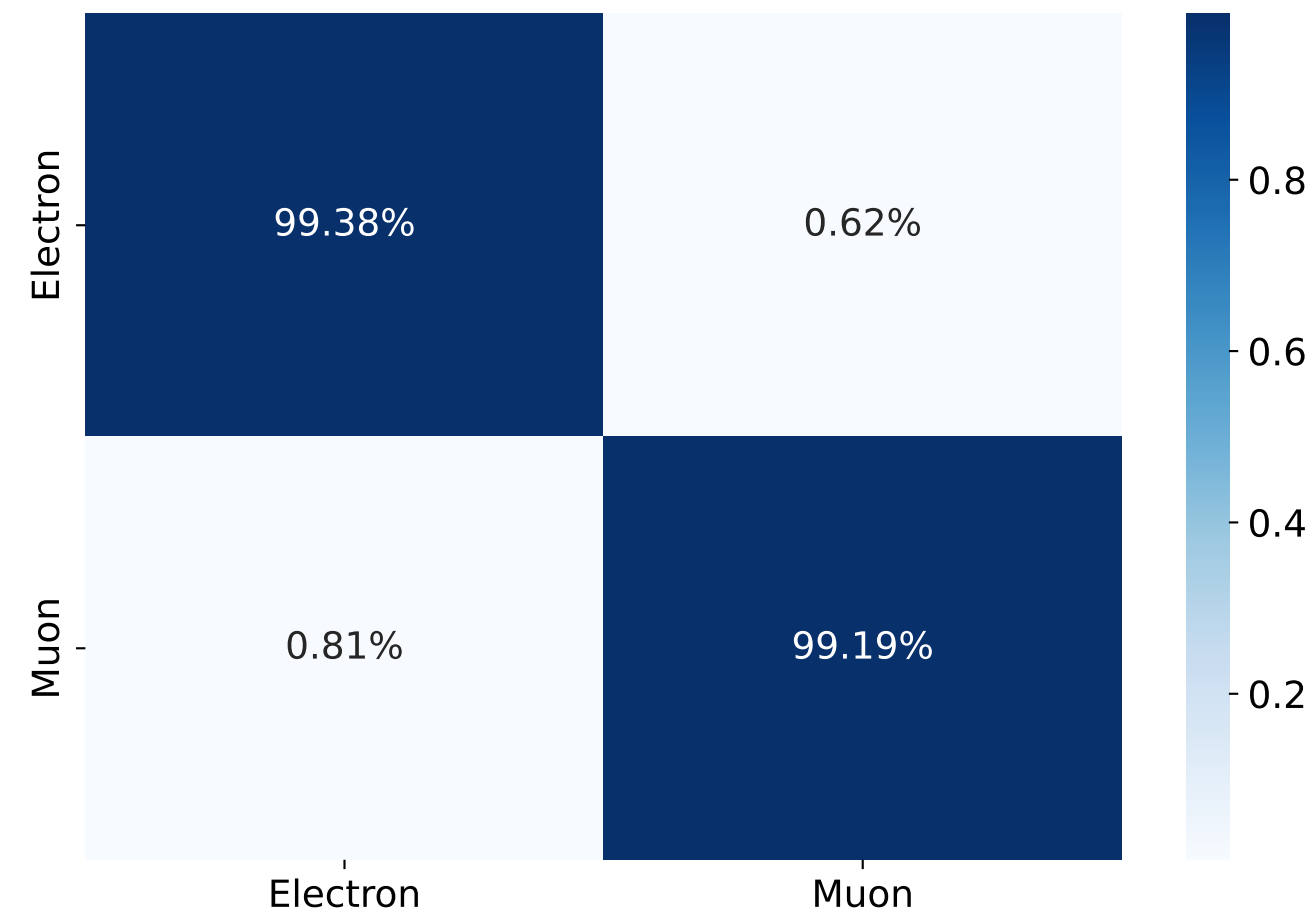
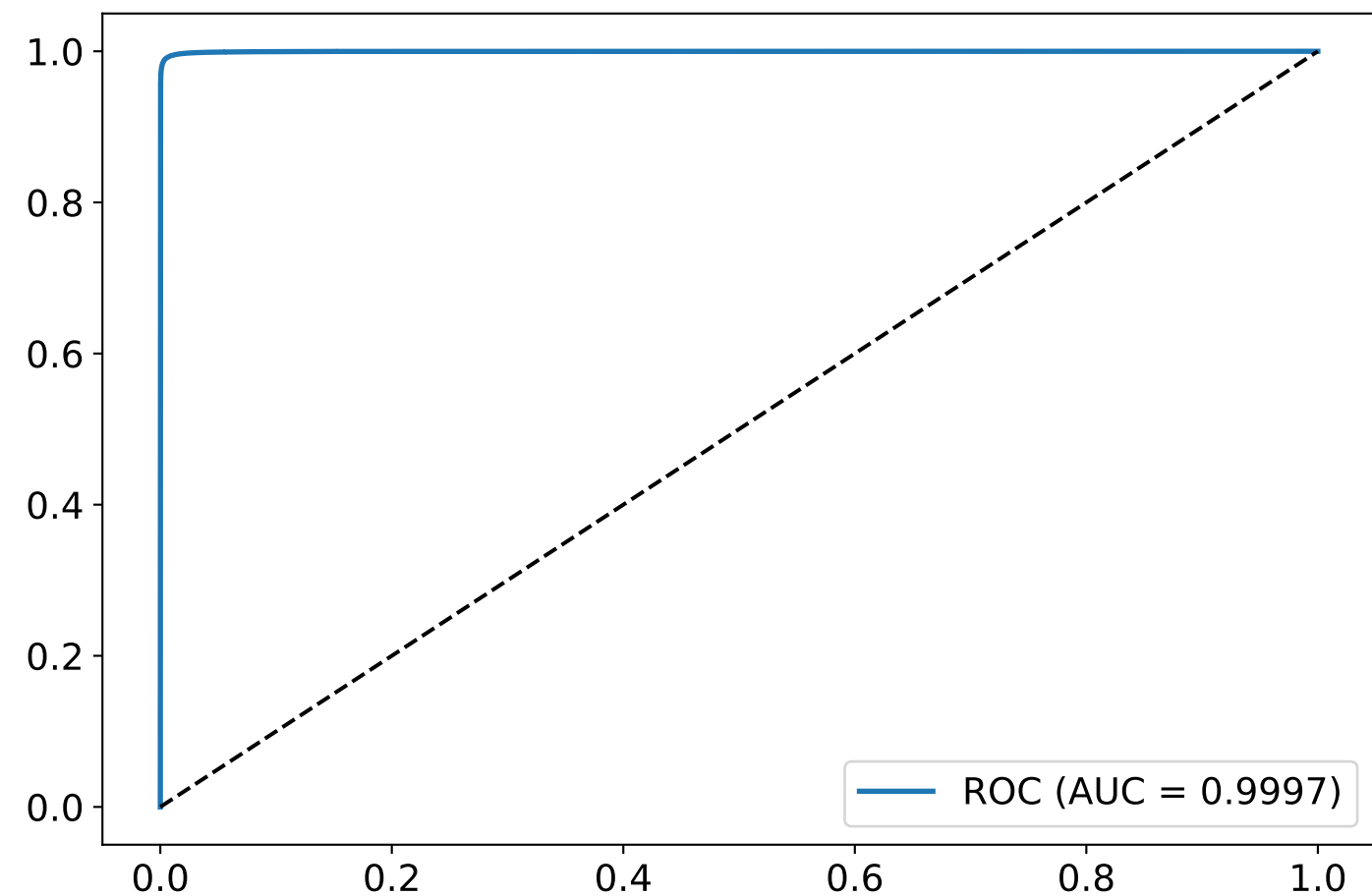


EtaRange = FullRange (TEST DATA) - Accuracy: 0.992

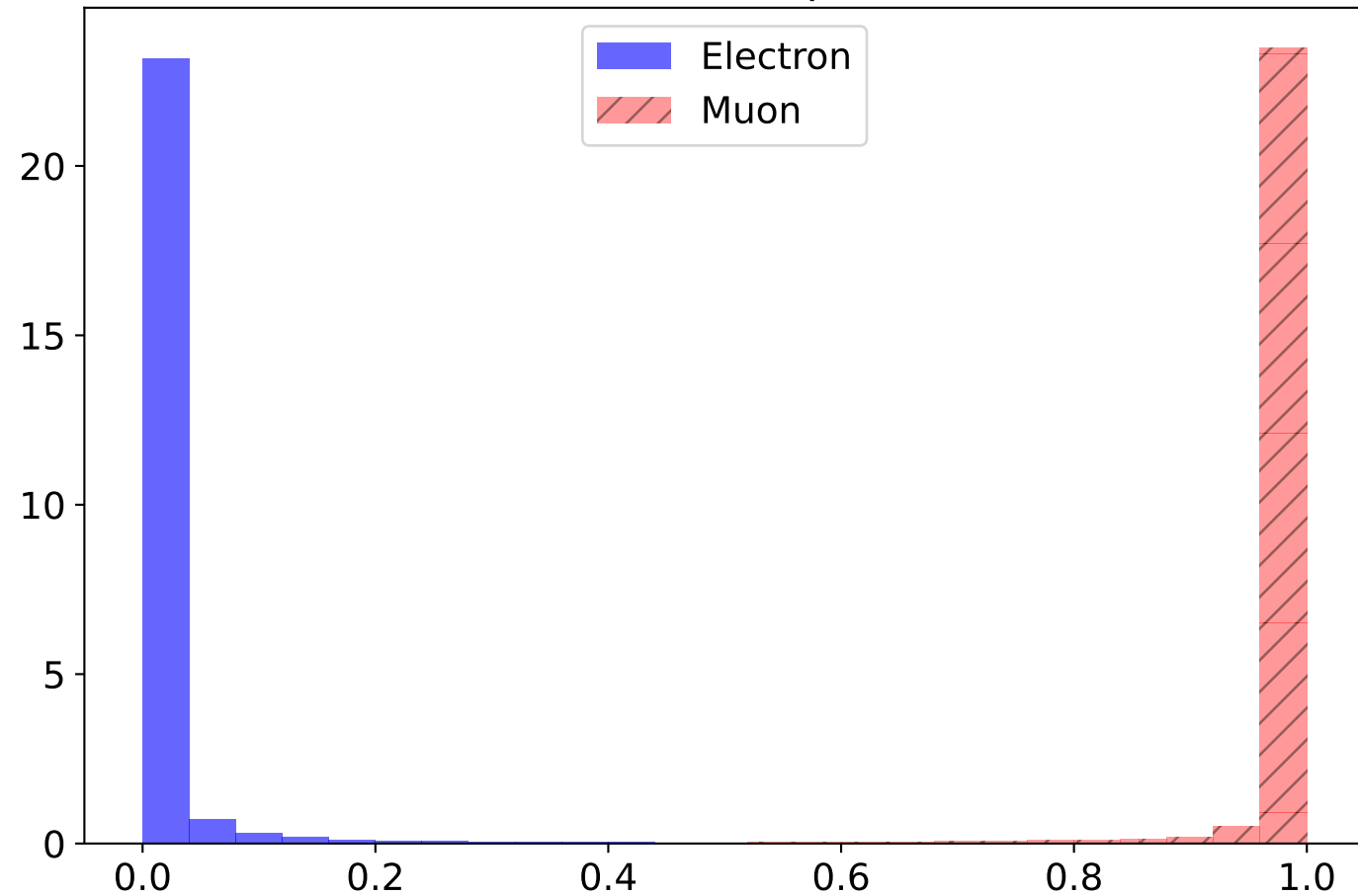
Confusion Matrix



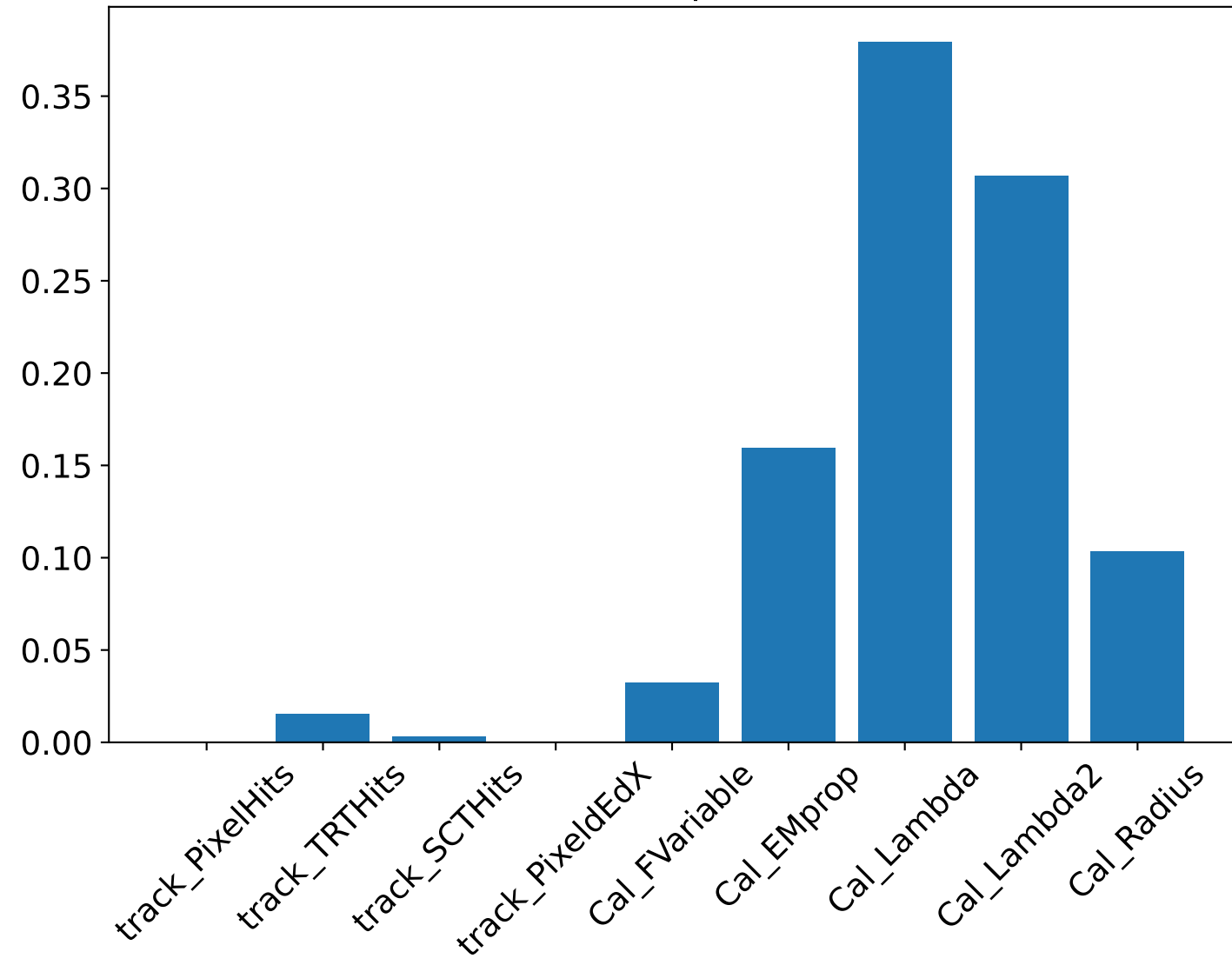
ROC Curve



Classifier Response

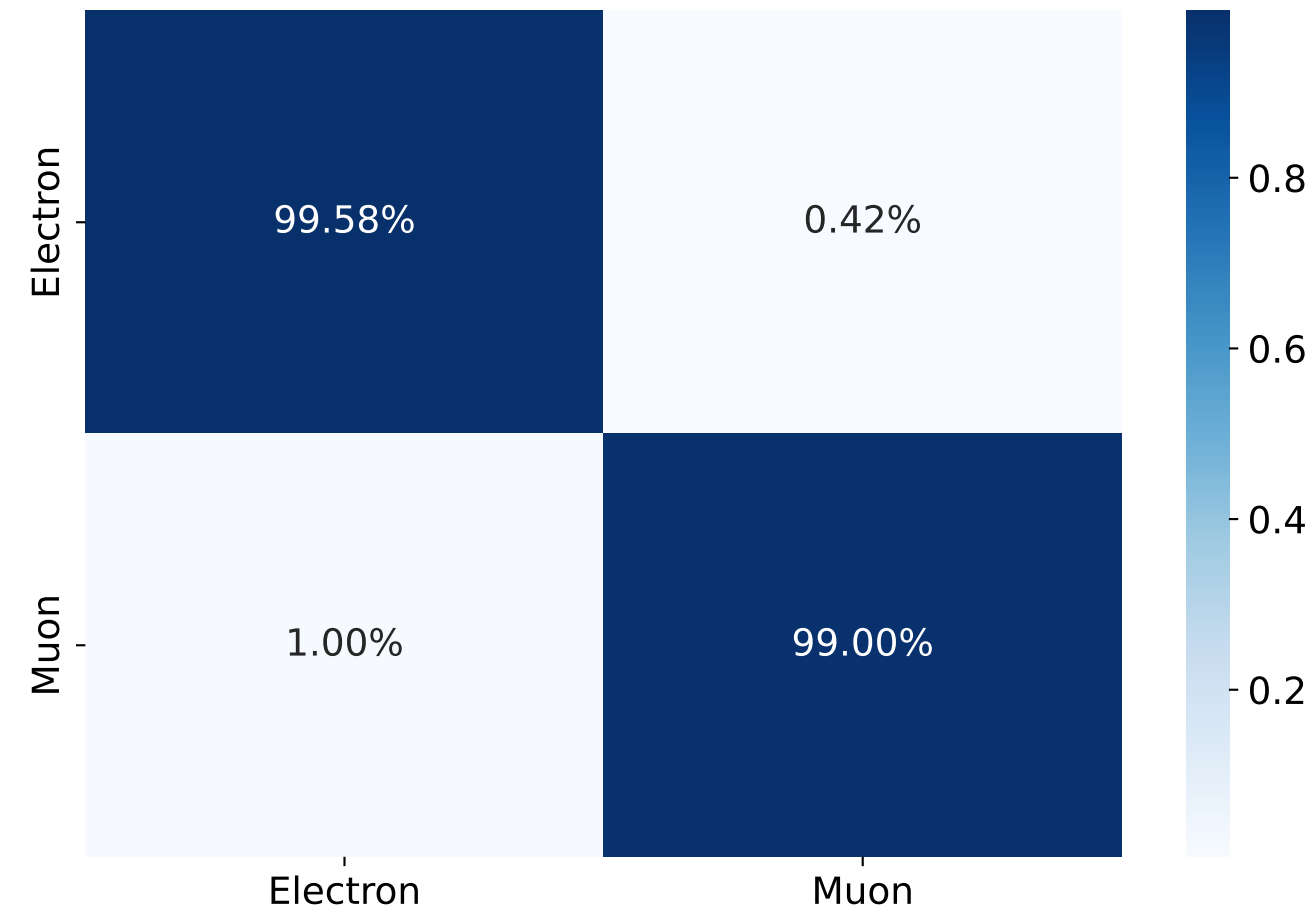


Feature Importance

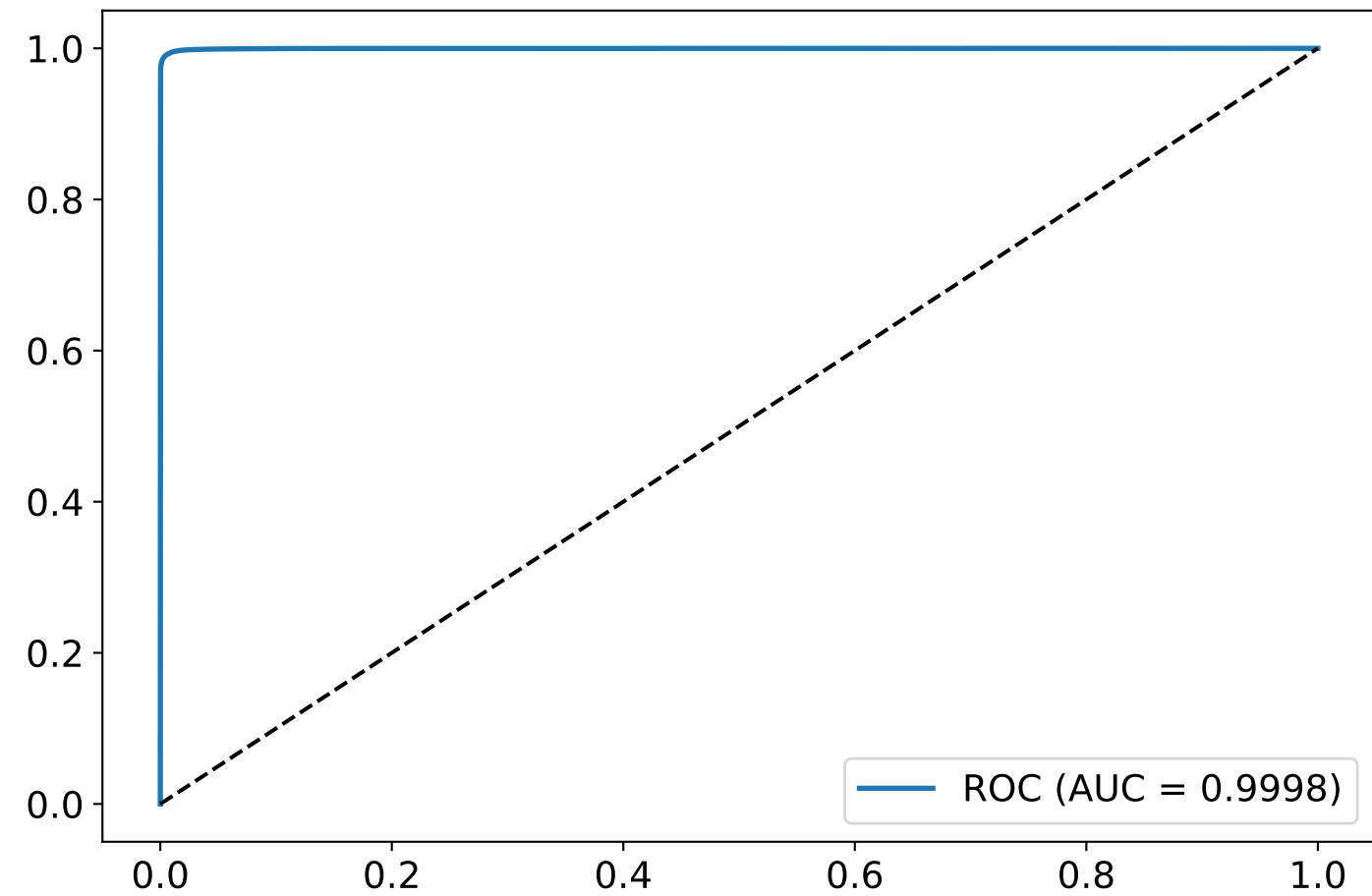


EtaRange = <1.0 (TEST DATA) - Accuracy: 0.992

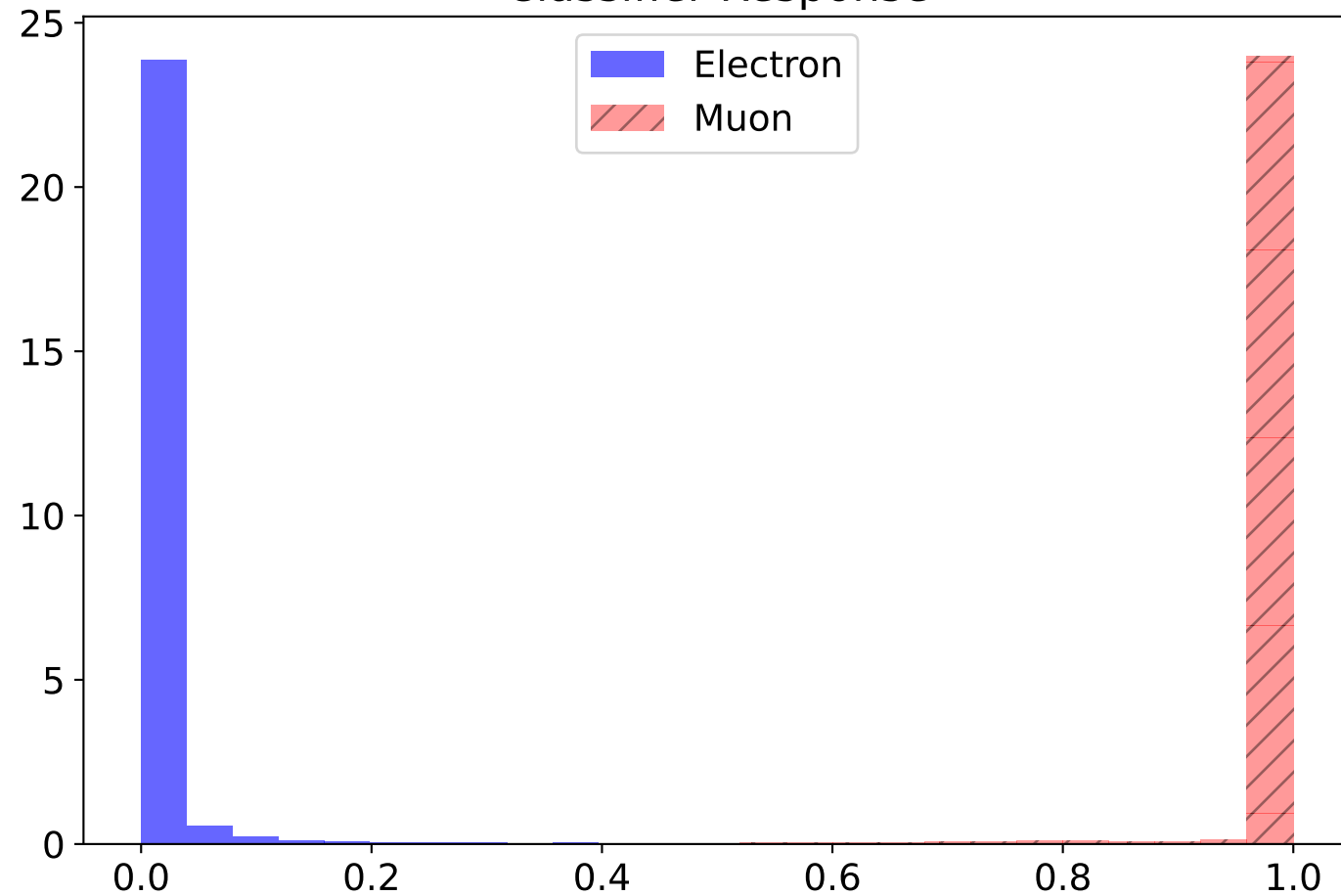
Confusion Matrix



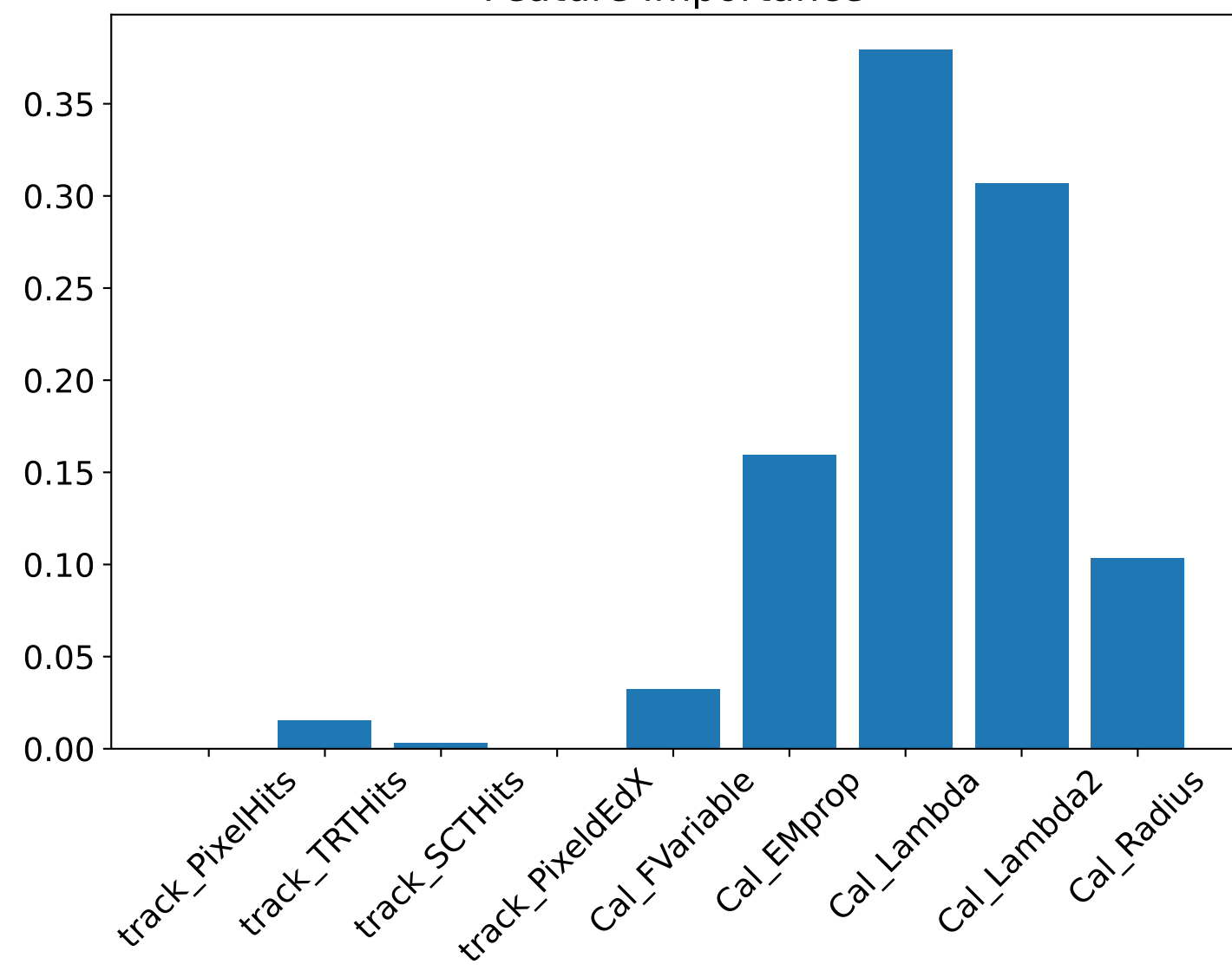
ROC Curve



Classifier Response

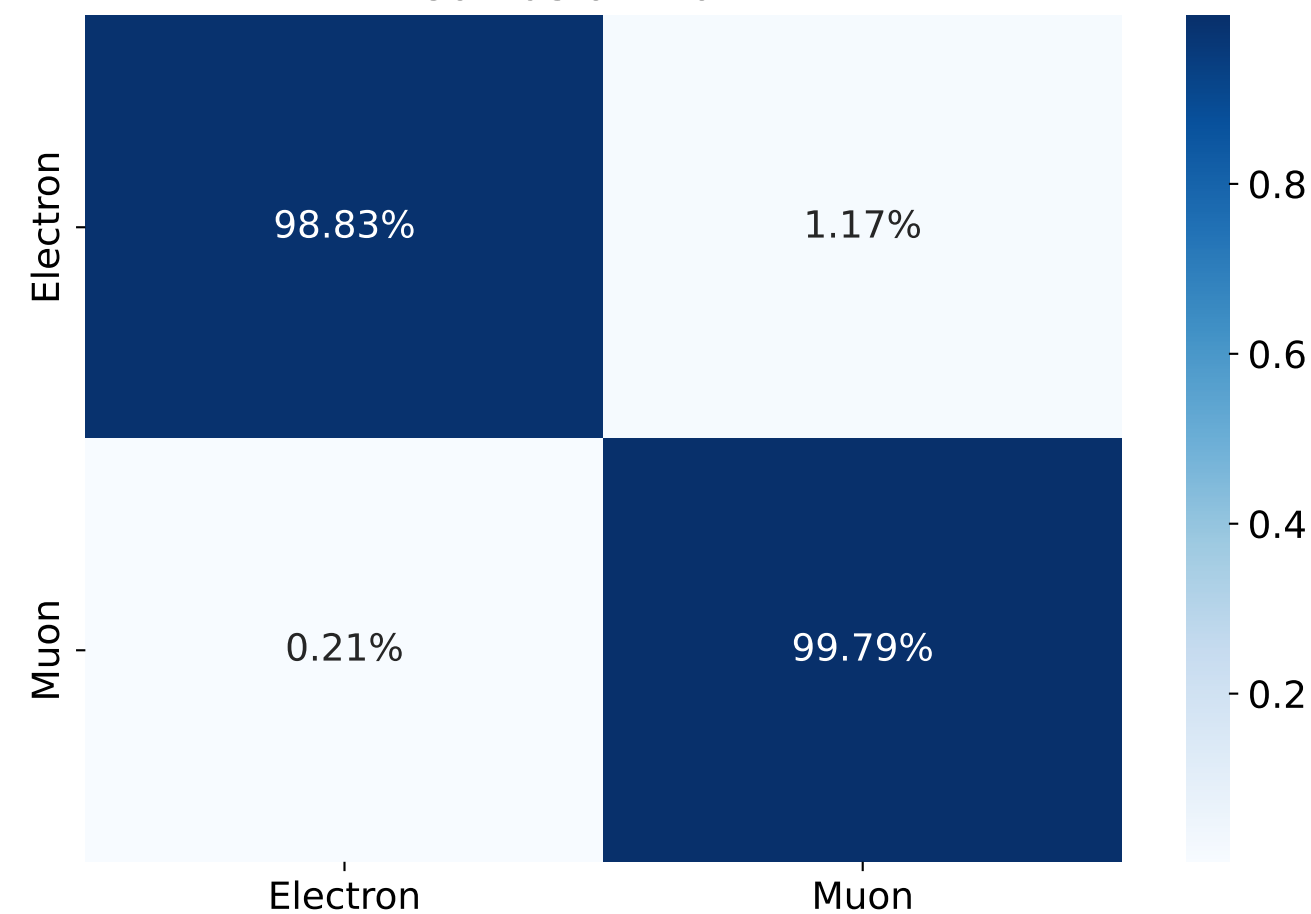


Feature Importance

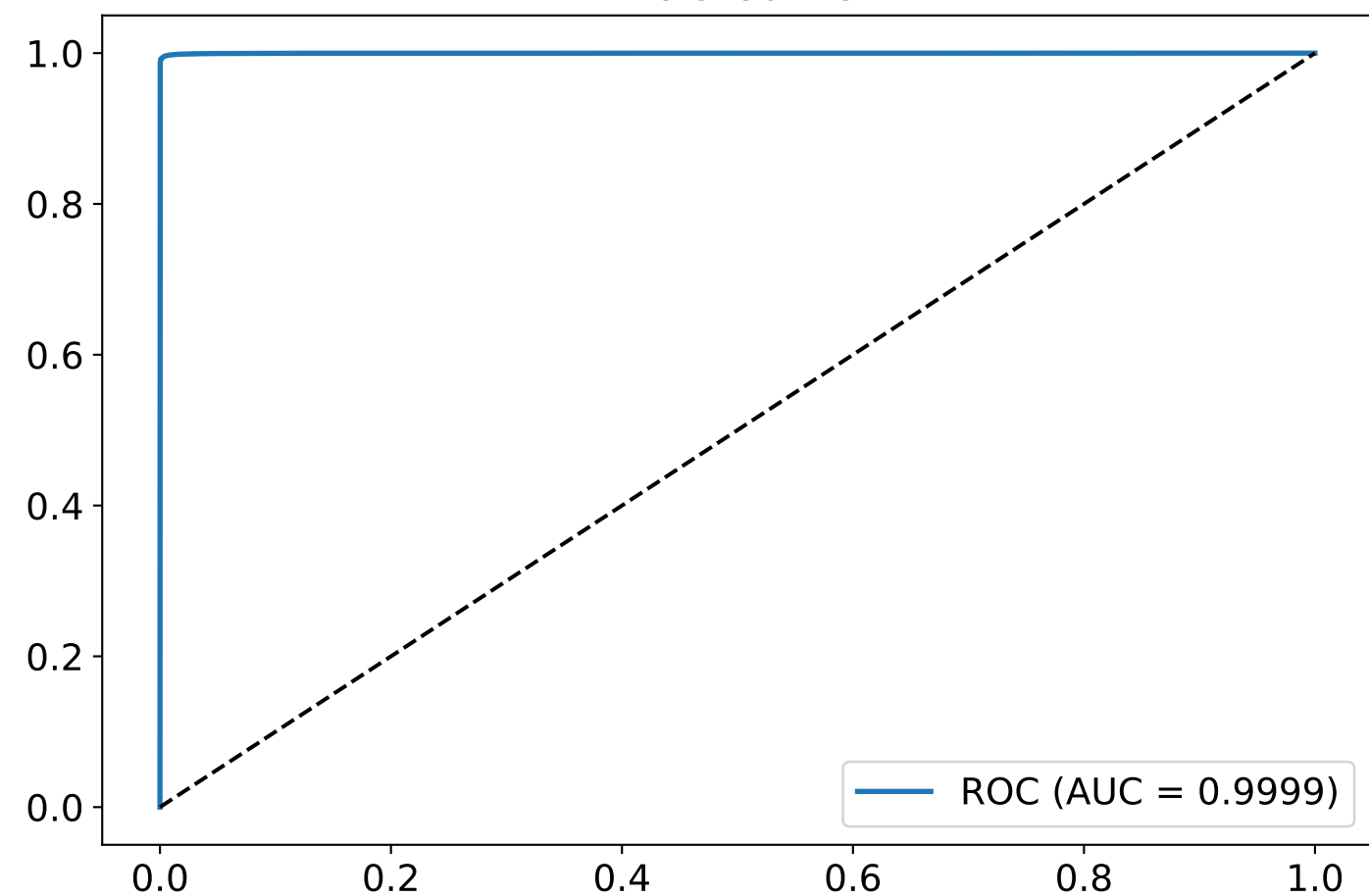


EtaRange = 1.0-1.5 (TEST DATA) - Accuracy: 0.996

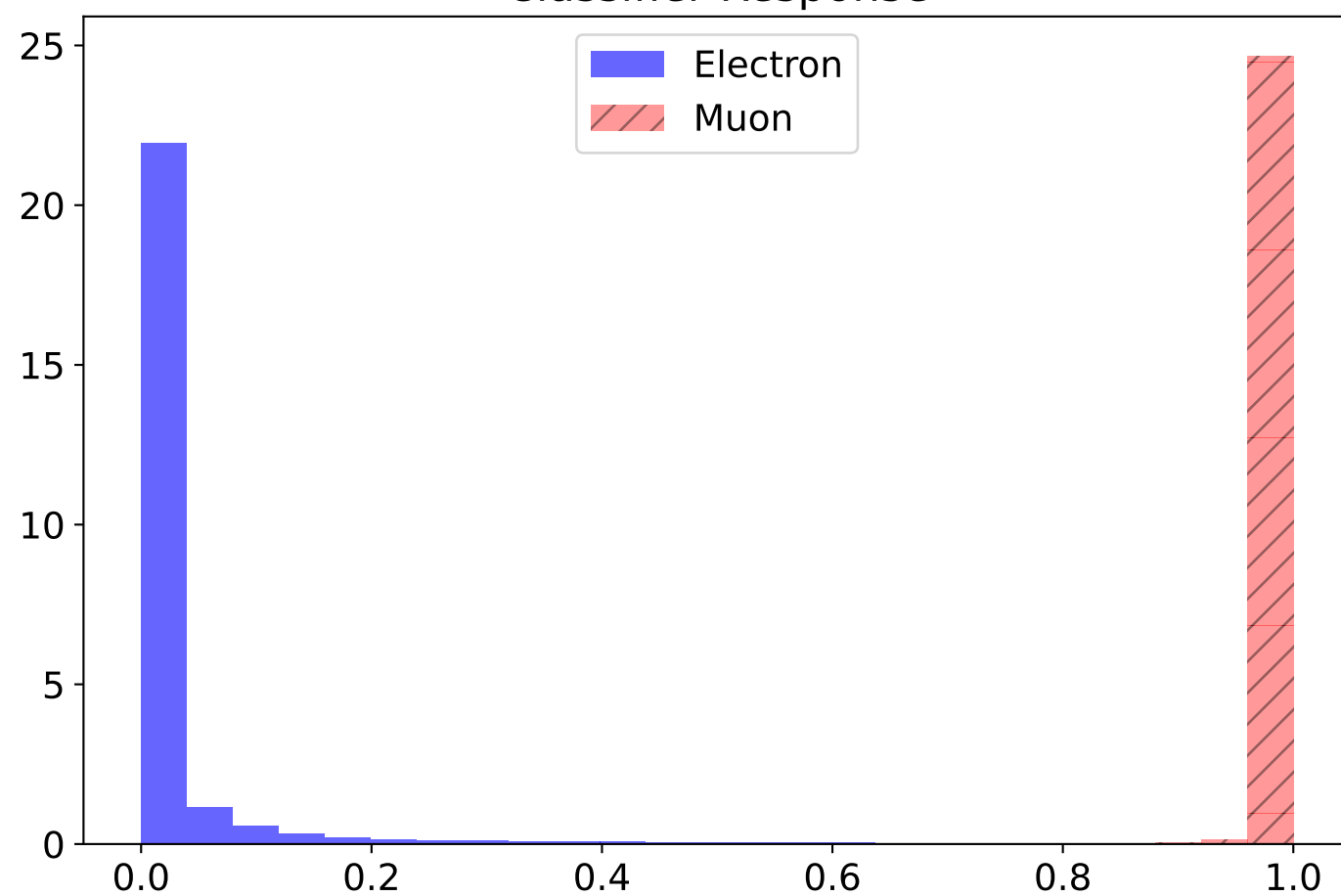
Confusion Matrix



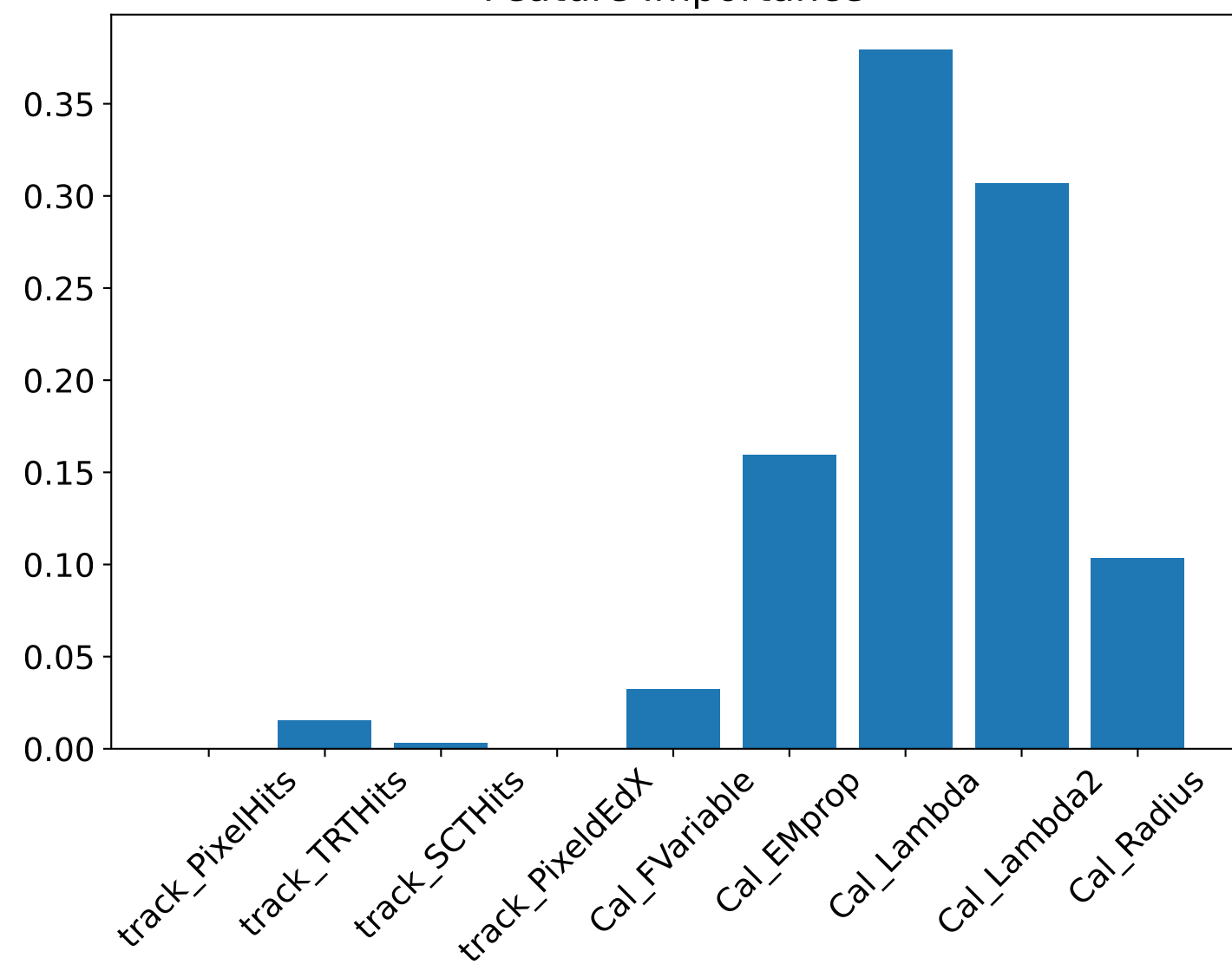
ROC Curve



Classifier Response

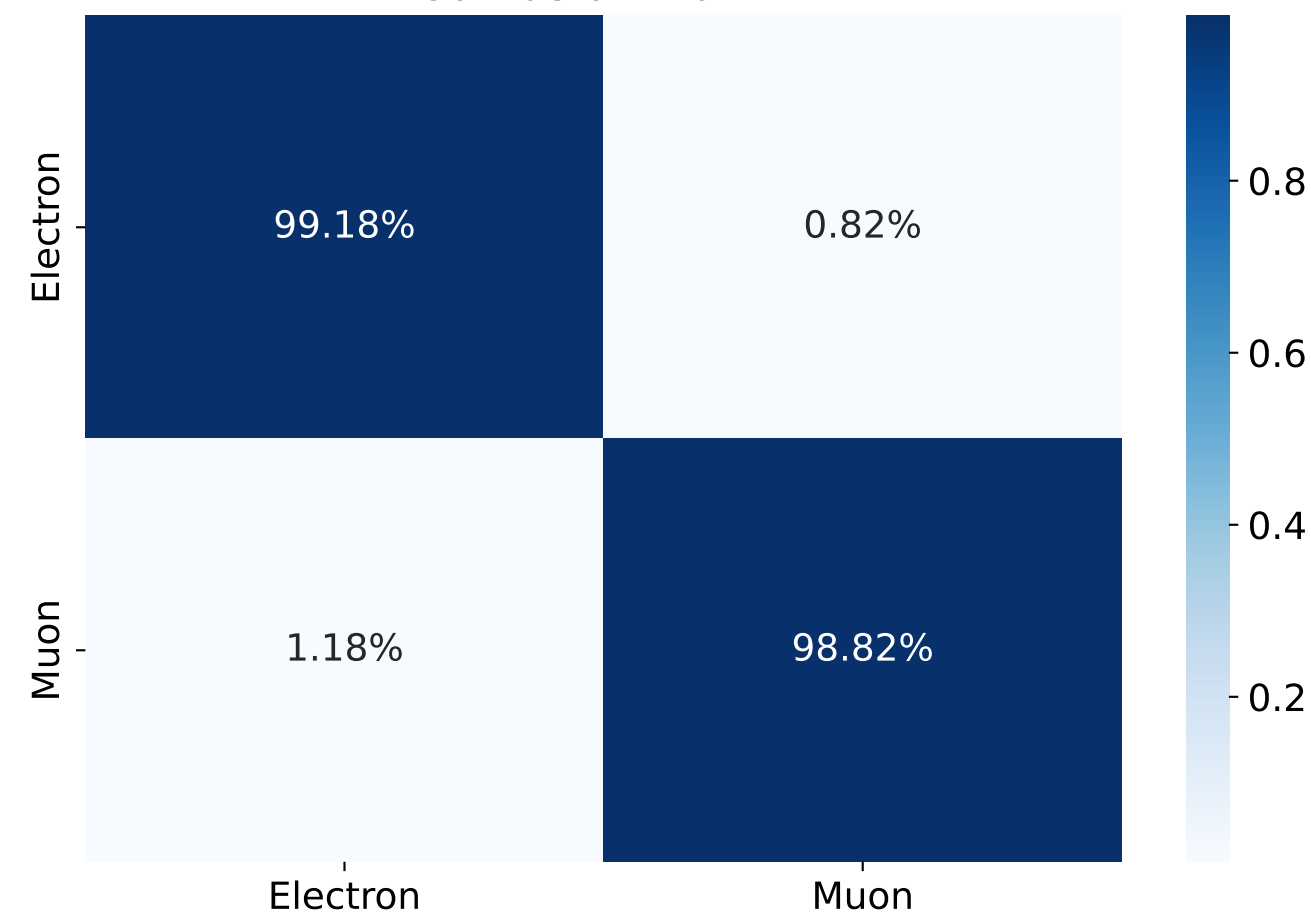


Feature Importance

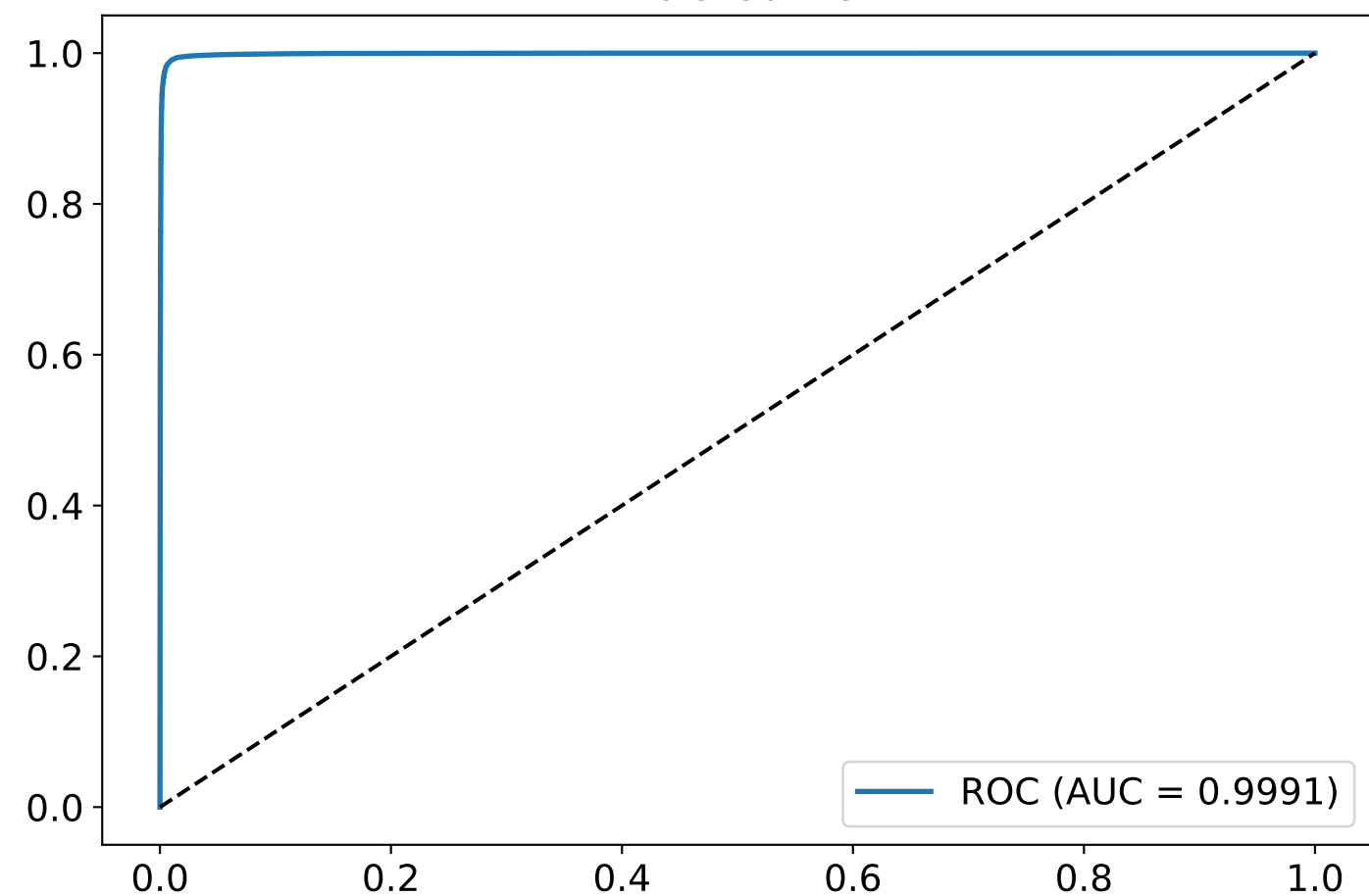


EtaRange = 1.5-2.5 (TEST DATA) - Accuracy: 0.989

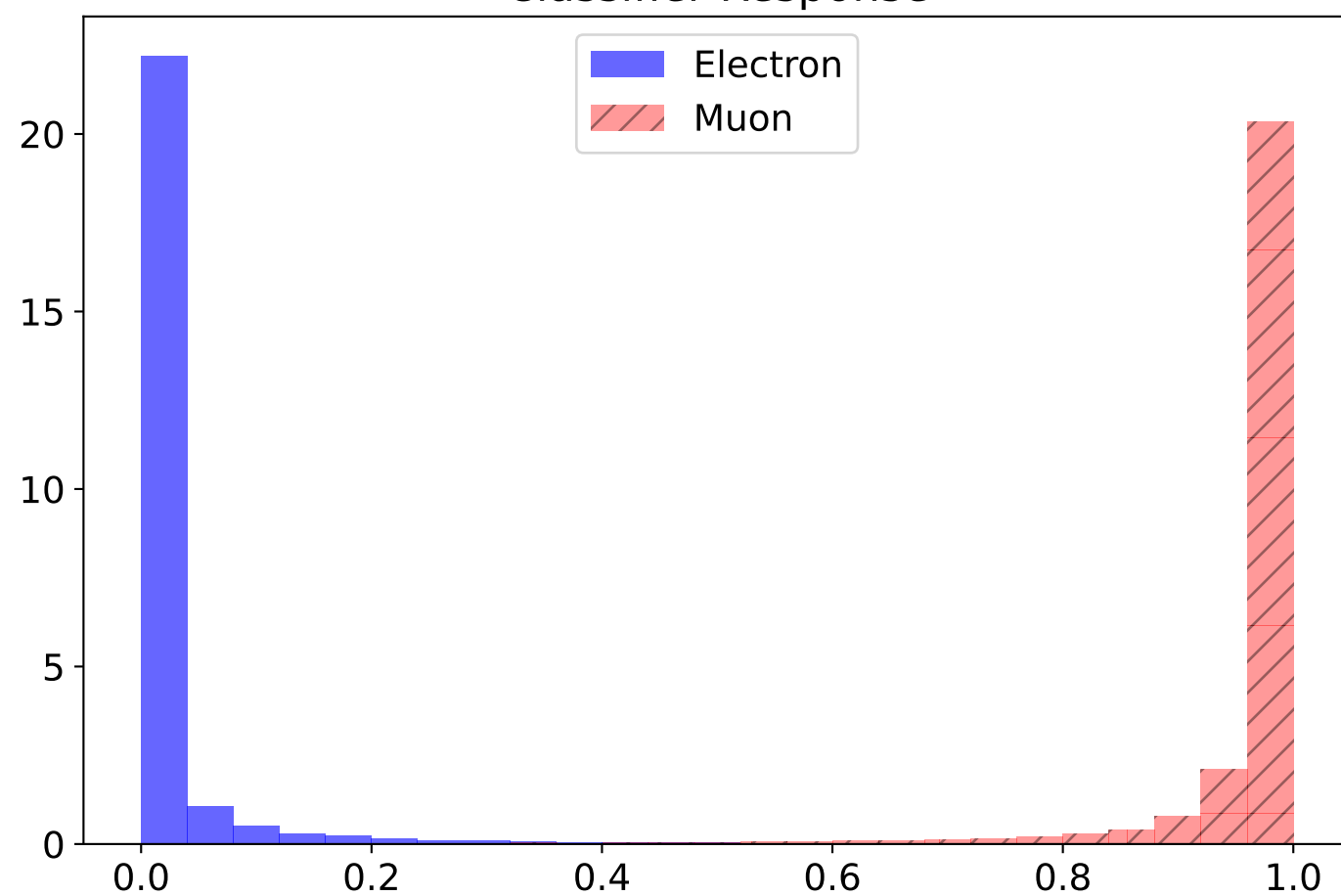
Confusion Matrix



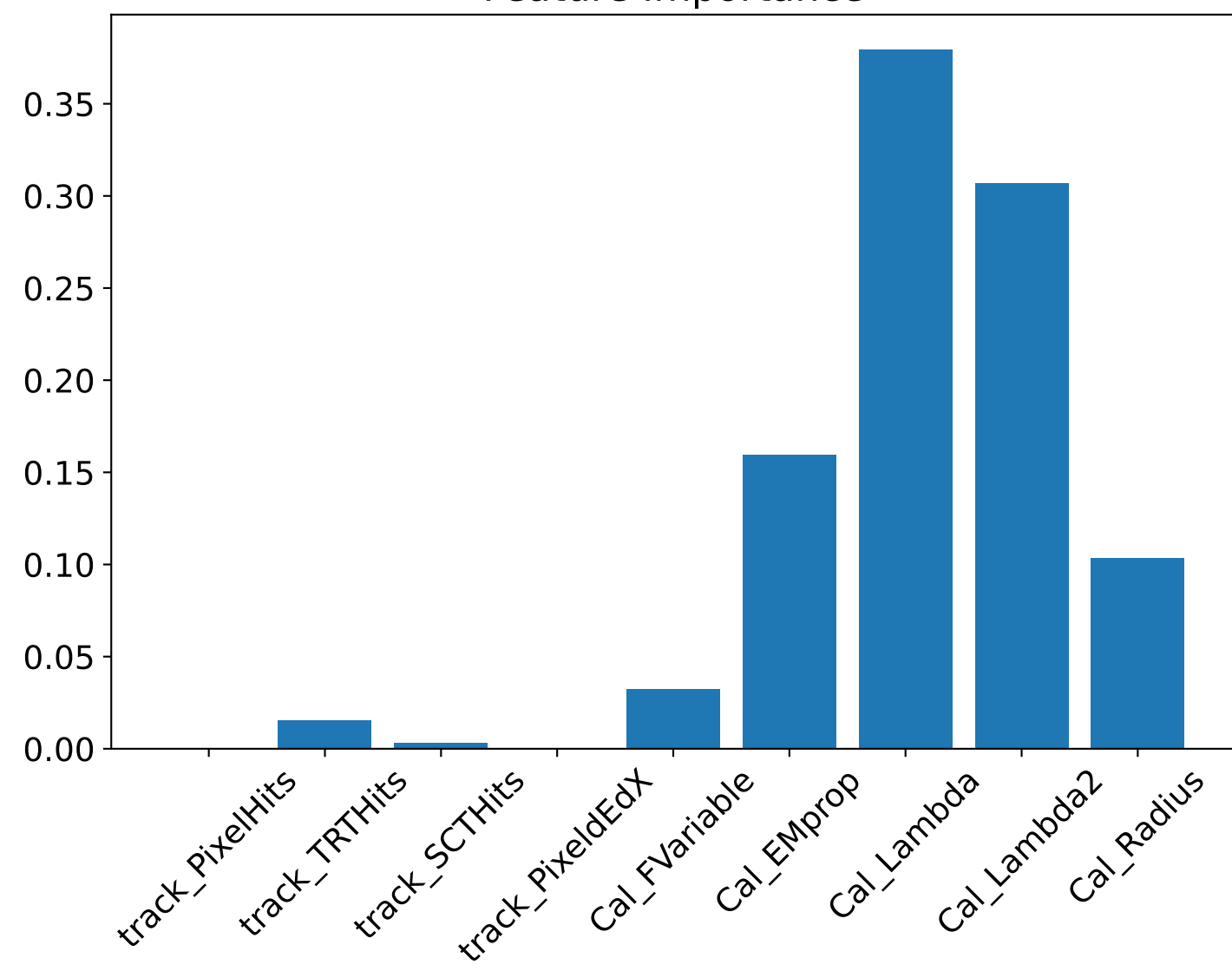
ROC Curve

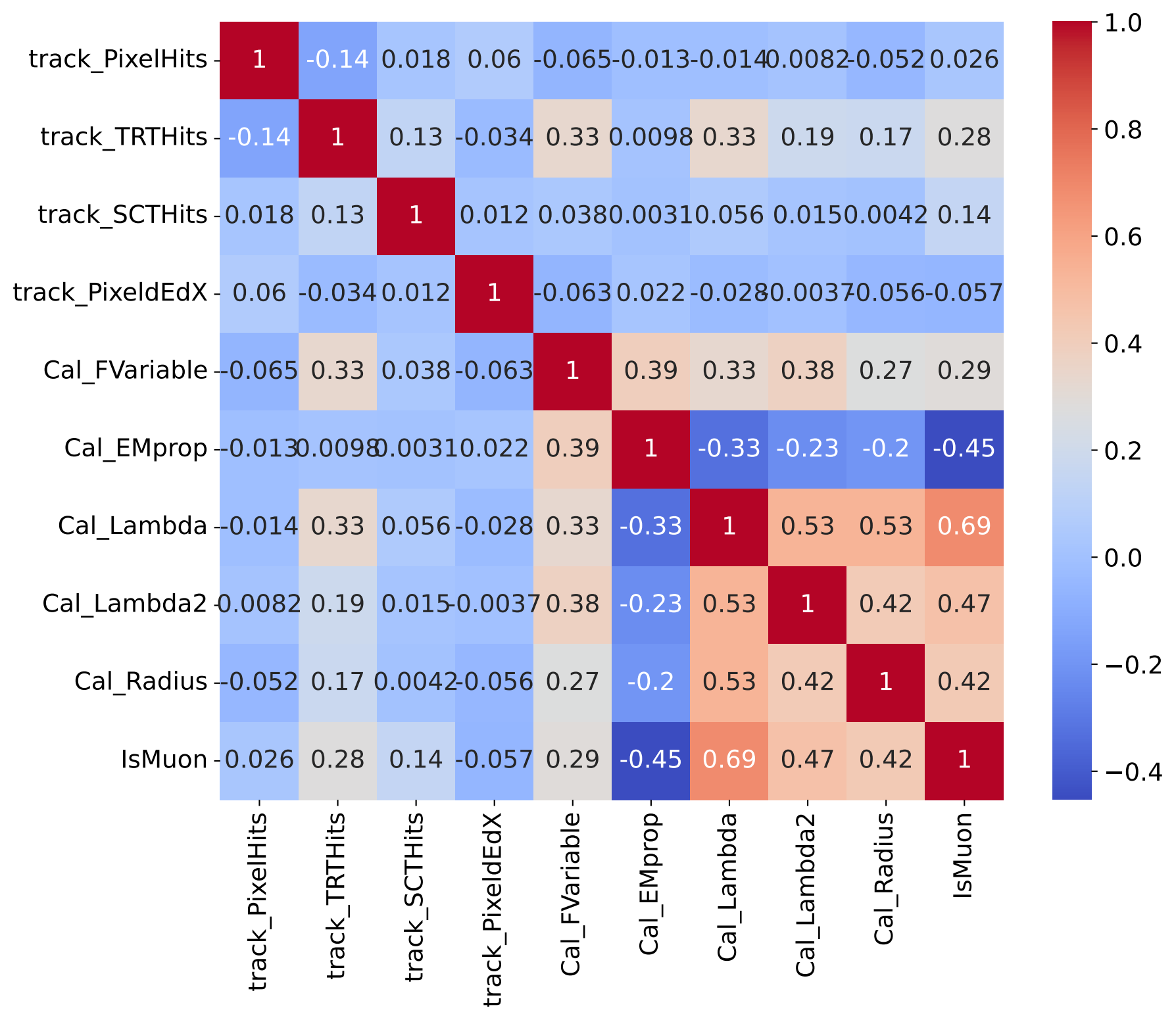


Classifier Response



Feature Importance





Classification Reports Summary

Trained: MCData.root | Test: MCData.root

EtaRange = FullRange

	precision	recall	f1-score	support
Electron	0.98	0.99	0.99	158732
Muon	1.00	0.99	0.99	439792
accuracy			0.99	598524
macro avg	0.99	0.99	0.99	598524
weighted avg	0.99	0.99	0.99	598524

EtaRange = <1.0

	precision	recall	f1-score	support
Electron	0.98	1.00	0.99	106881
Muon	1.00	0.99	0.99	234767
accuracy			0.99	341648
macro avg	0.99	0.99	0.99	341648
weighted avg	0.99	0.99	0.99	341648

EtaRange = 1.0-1.5

	precision	recall	f1-score	support
Electron	0.99	0.99	0.99	31172
Muon	1.00	1.00	1.00	123450
accuracy			1.00	154622
macro avg	0.99	0.99	0.99	154622
weighted avg	1.00	1.00	1.00	154622

EtaRange = 1.5-2.5

	precision	recall	f1-score	support
Electron	0.96	0.99	0.97	20679
Muon	1.00	0.99	0.99	81575
accuracy			0.99	102254
macro avg	0.98	0.99	0.98	102254
weighted avg	0.99	0.99	0.99	102254